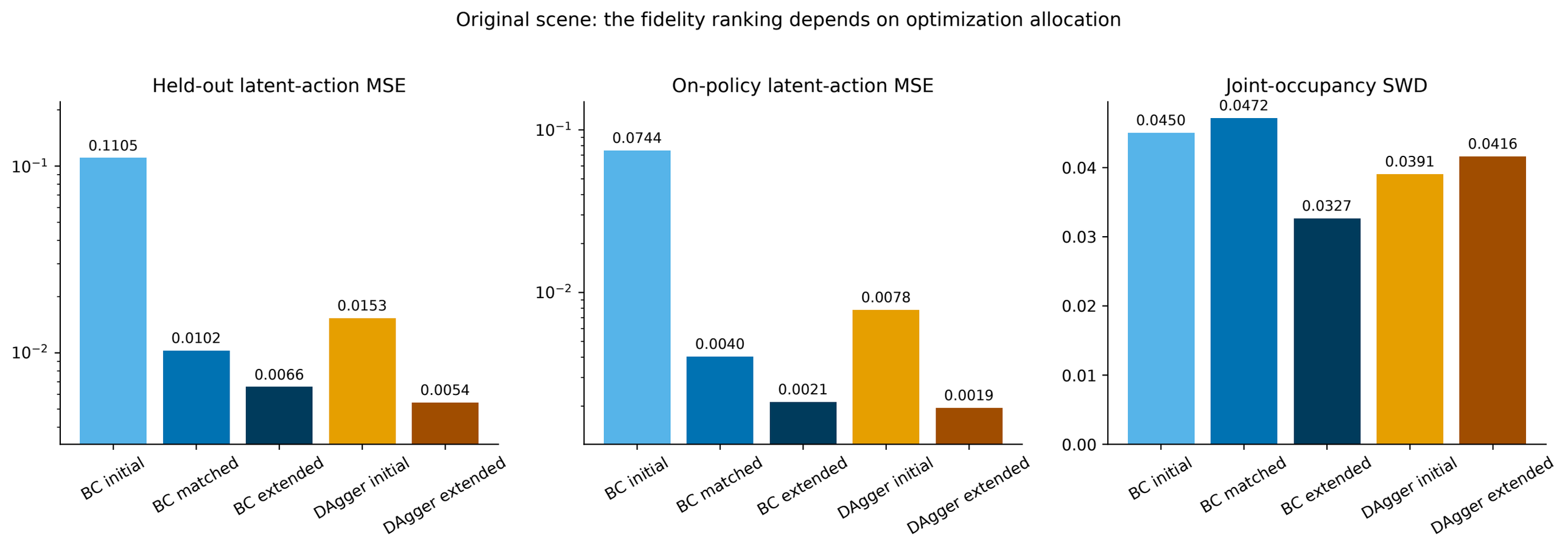
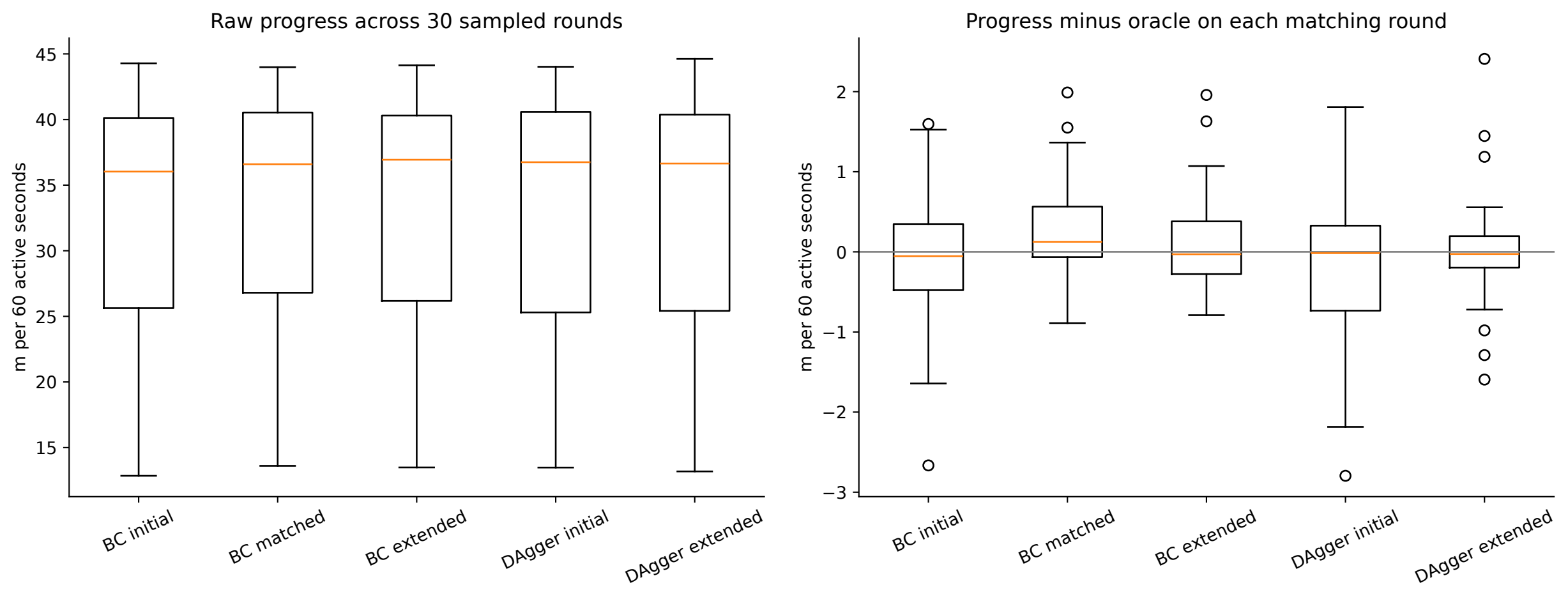
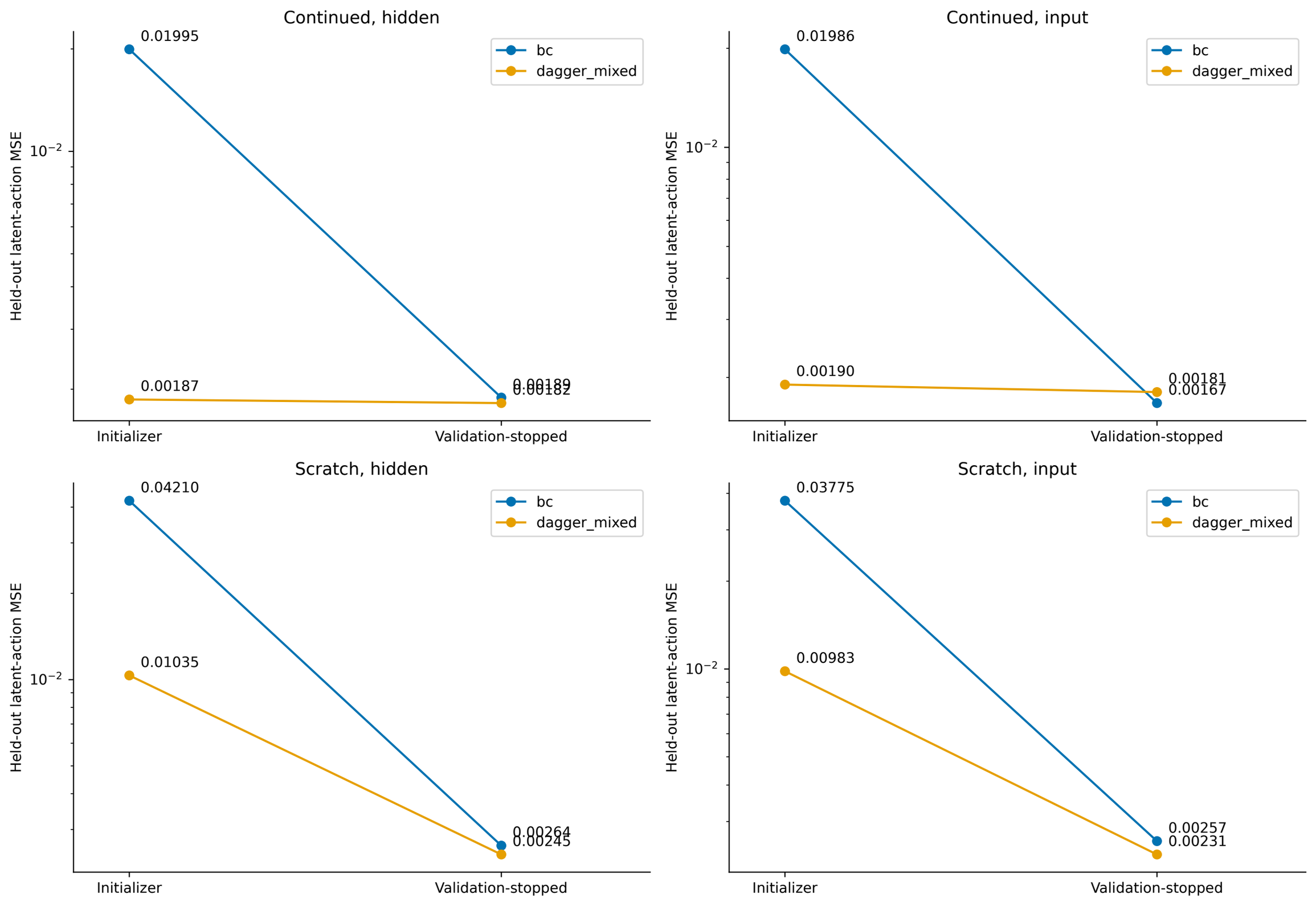
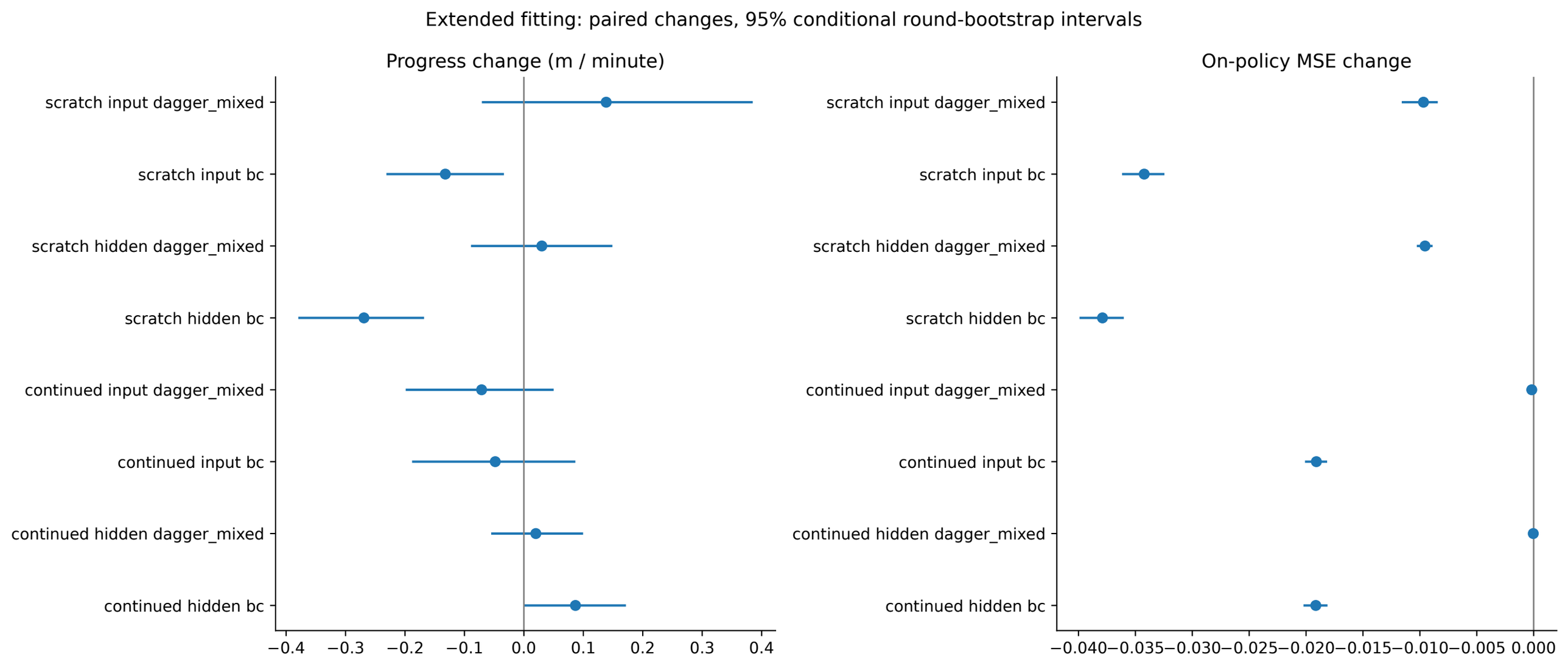
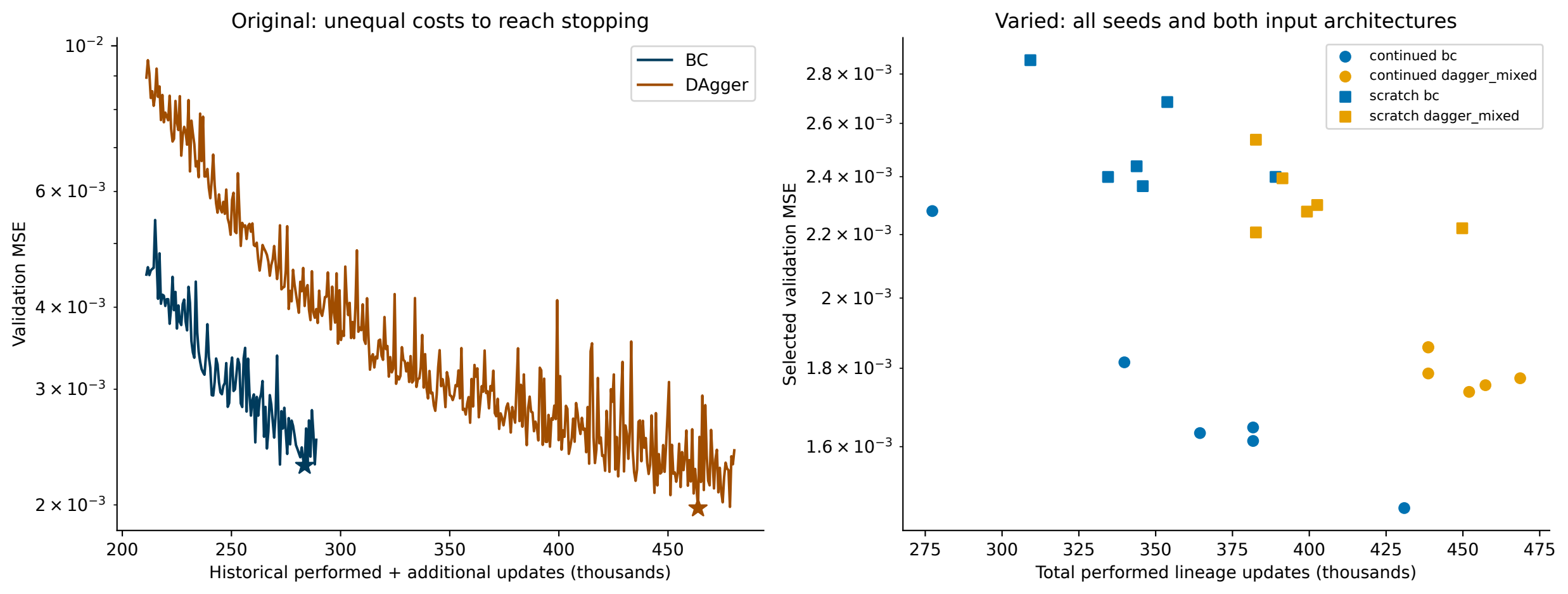
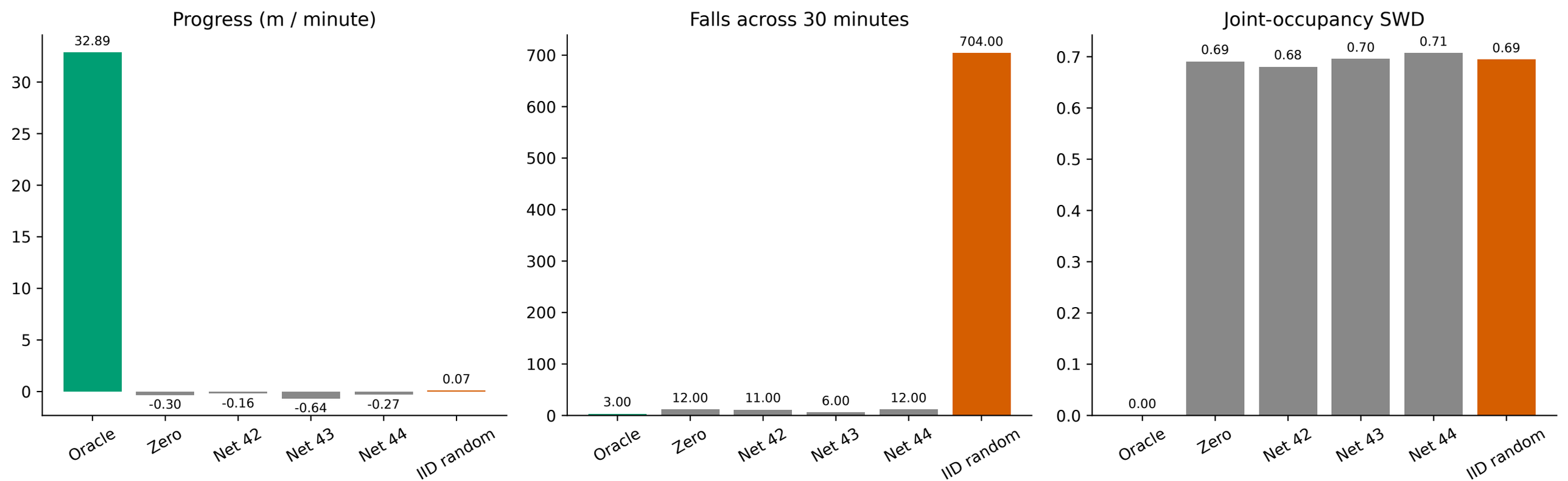

Original scene: the fidelity ranking depends on optimization allocation
Held-out latent-action MSE
10^{-1}
10^{-2}
0.1105
0.0102
0.0066
0.0153
0.0054
BC initial
BC matched
BC extended
DAgger initial
DAgger extended
On-policy latent-action MSE
10^{-1}
10^{-2}
0.0744
0.0040
0.0021
0.0078
0.0019
BC initial
BC matched
BC extended
DAgger initial
DAgger extended
Joint-occupancy SWD
0.04
0.03
0.02
0.01
0.00
0.0450
0.0472
0.0327
0.0391
0.0416
BC initial
BC matched
BC extended
DAgger initial
DAgger extended

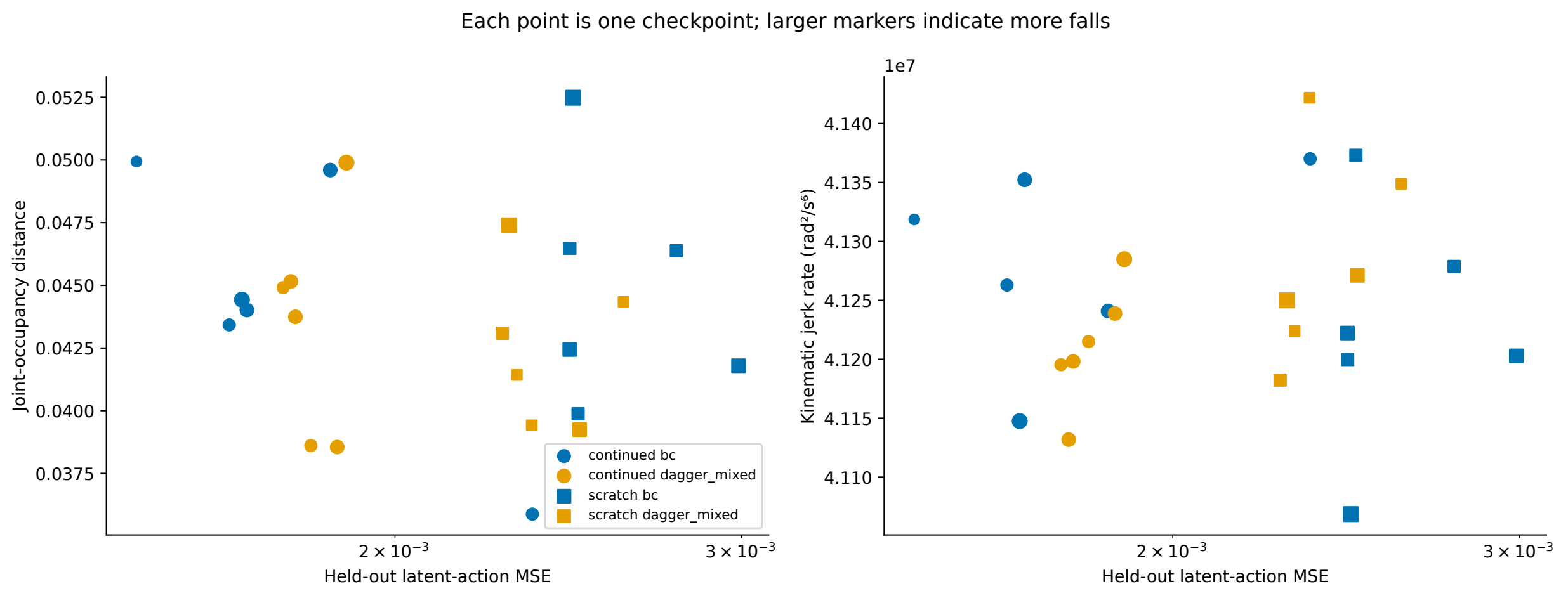

Raw progress across 30 sampled rounds
m per 60 active seconds
45
40
35
30
25
20
15
BC initial
BC matched
BC extended
DAgger initial
DAgger extended
Progress minus oracle on each matching round
m per 60 active seconds
2
1
0
−1
−2
−3
BC initial
BC matched
BC extended
DAgger initial
DAgger extended

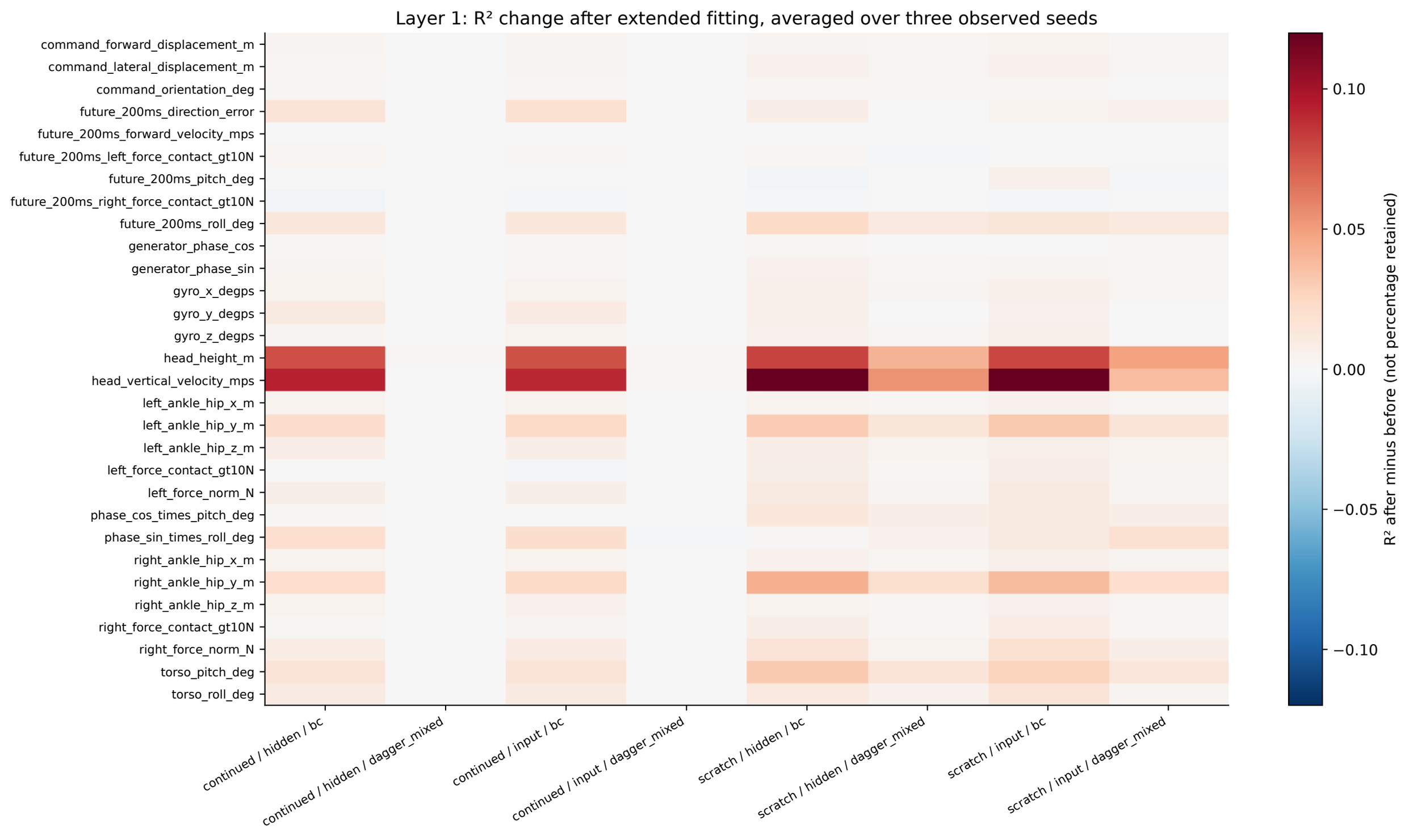

Continued, hidden
Held-out latent-action MSE
10^{-2}
bc
dagger_mixed
0.01995
0.00187
0.00189
0.00182
Initializer
Validation-stopped
Continued, input
Held-out latent-action MSE
10^{-2}
bc
dagger_mixed
0.01986
0.00190
0.00181
0.00167
Initializer
Validation-stopped
Scratch, hidden
Held-out latent-action MSE
10^{-2}
bc
dagger_mixed
0.04210
0.01035
0.00264
0.00245
Initializer
Validation-stopped
Scratch, input
Held-out latent-action MSE
10^{-2}
bc
dagger_mixed
0.03775
0.00983
0.00257
0.00231
Initializer
Validation-stopped

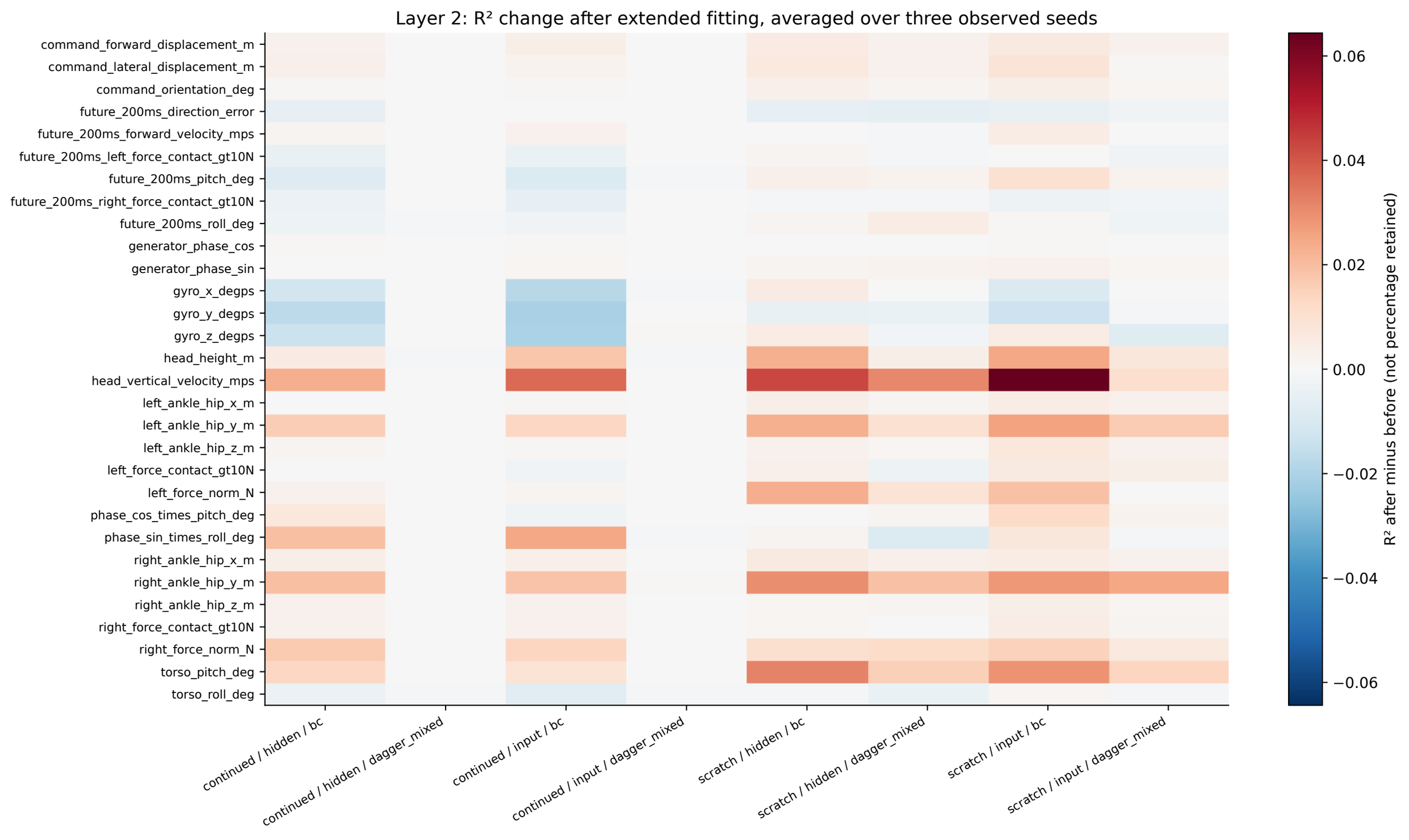

Extended fitting: paired changes, 95% conditional round-bootstrap intervals
Progress change (m / minute)
On-policy MSE change
scratch input dagger_mixed
scratch input bc
scratch hidden dagger_mixed
scratch hidden bc
continued input dagger_mixed
continued input bc
continued hidden dagger_mixed
continued hidden bc
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
−0.040
−0.035
−0.030
−0.025
−0.020
−0.015
−0.010
−0.005
0.000

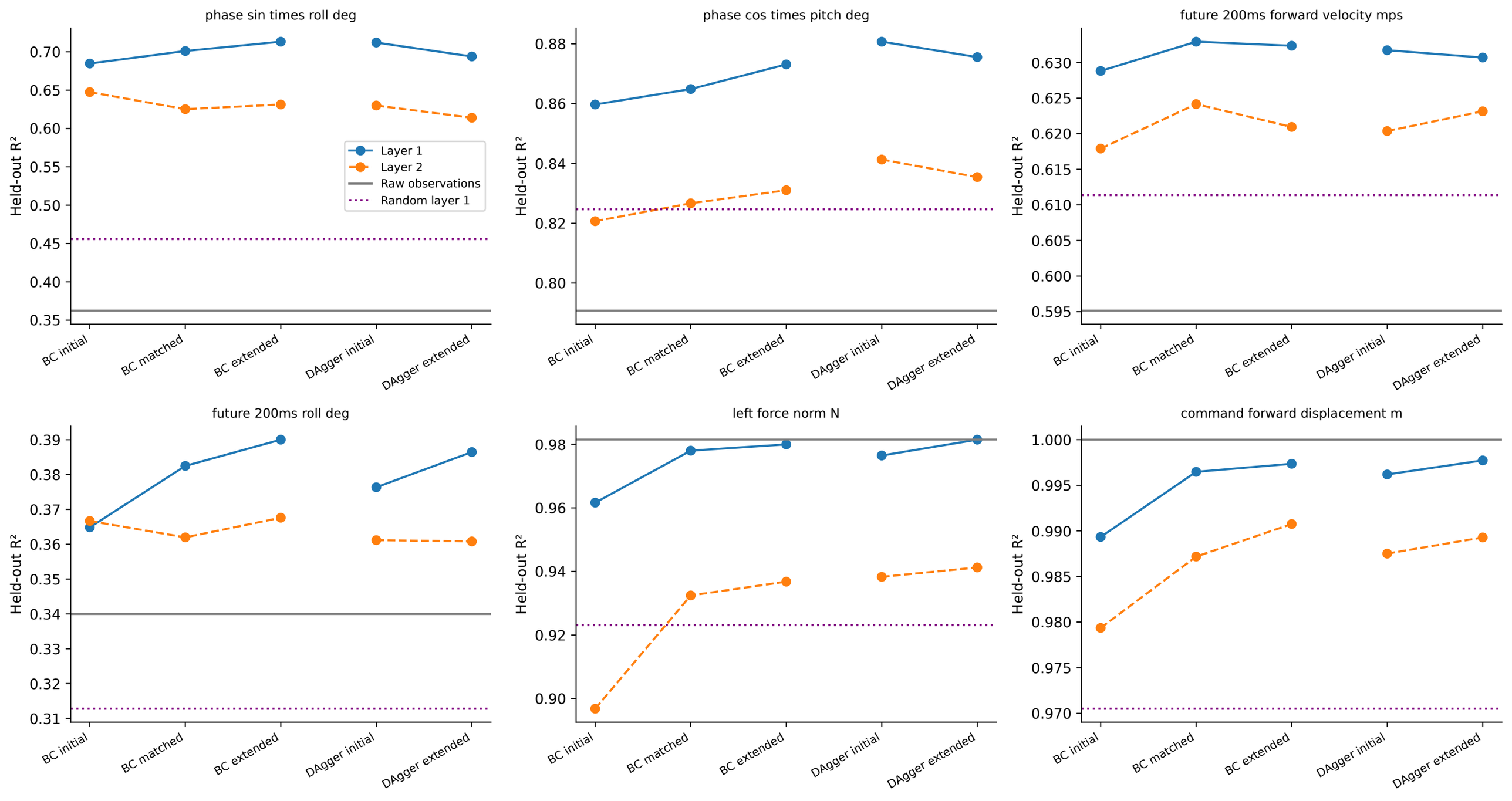

Original: unequal costs to reach stopping
BC
DAgger
Validation MSE
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
200
250
300
350
400
450
Historical performed + additional updates (thousands)
Varied: all seeds and both input architectures
continued bc
continued dagger_mixed
scratch bc
scratch dagger_mixed
Selected validation MSE
2.8×10^{-3}
2.6×10^{-3}
2.4×10^{-3}
2.2×10^{-3}
2×10^{-3}
1.8×10^{-3}
1.6×10^{-3}
275
300
325
350
375
400
425
450
475
Total performed lineage updates (thousands)

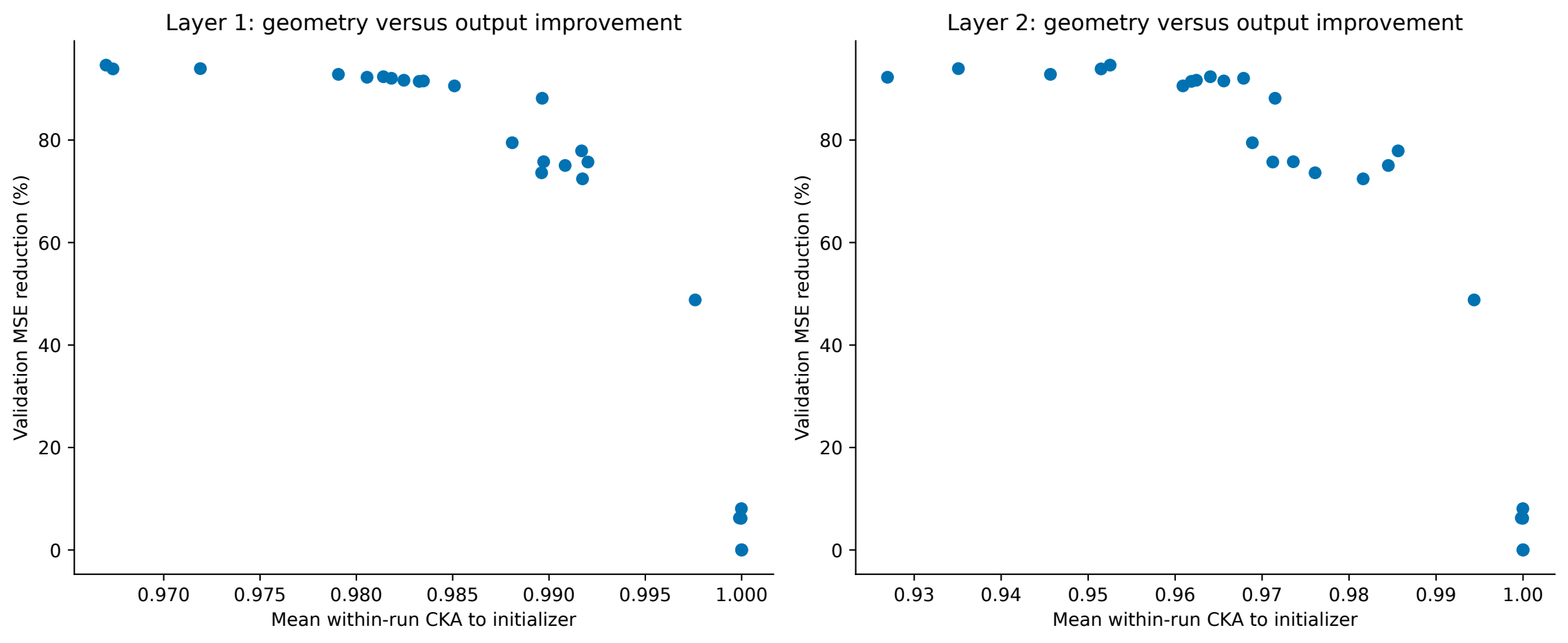

Progress (m / minute)
32.89
-0.30
-0.16
-0.64
-0.27
0.07
Oracle
Zero
Net 42
Net 43
Net 44
IID random
Falls across 30 minutes
704.00
3.00
12.00
11.00
6.00
12.00
Joint-occupancy SWD
0.00
0.69
0.68
0.70
0.71
0.69


Each point is one checkpoint; larger markers indicate more falls
Joint-occupancy distance
0.0525
0.0500
0.0475
0.0450
0.0425
0.0400
0.0375
2×10^{-3}
3×10^{-3}
Held-out latent-action MSE
continued bc
continued dagger_mixed
scratch bc
scratch dagger_mixed
1e7
Kinematic jerk rate (rad²/s⁶)
4.140
4.135
4.130
4.125
4.120
4.115
4.110
2×10^{-3}
3×10^{-3}
Held-out latent-action MSE


Layer 1: R² change after extended fitting, averaged over three observed seeds
command_forward_displacement_m
command_lateral_displacement_m
command_orientation_deg
future_200ms_direction_error
future_200ms_forward_velocity_mps
future_200ms_left_force_contact_gt10N
future_200ms_pitch_deg
future_200ms_right_force_contact_gt10N
future_200ms_roll_deg
generator_phase_cos
generator_phase_sin
gyro_x_degps
gyro_y_degps
gyro_z_degps
head_height_m
head_vertical_velocity_mps
left_ankle_hip_x_m
left_ankle_hip_y_m
left_ankle_hip_z_m
left_force_contact_gt10N
left_force_norm_N
phase_cos_times_pitch_deg
phase_sin_times_roll_deg
right_ankle_hip_x_m
right_ankle_hip_y_m
right_ankle_hip_z_m
right_force_contact_gt10N
right_force_norm_N
torso_pitch_deg
torso_roll_deg
continued / hidden / bc
continued / hidden / dagger_mixed
continued / input / bc
continued / input / dagger_mixed
scratch / hidden / bc
scratch / hidden / dagger_mixed
scratch / input / bc
scratch / input / dagger_mixed
0.10
0.05
0.00
−0.05
−0.10
R² after minus before (not percentage retained)


Layer 2: R² change after extended fitting, averaged over three observed seeds
command_forward_displacement_m
command_lateral_displacement_m
command_orientation_deg
future_200ms_direction_error
future_200ms_forward_velocity_mps
future_200ms_left_force_contact_gt10N
future_200ms_pitch_deg
future_200ms_right_force_contact_gt10N
future_200ms_roll_deg
generator_phase_cos
generator_phase_sin
gyro_x_degps
gyro_y_degps
gyro_z_degps
head_height_m
head_vertical_velocity_mps
left_ankle_hip_x_m
left_ankle_hip_y_m
left_ankle_hip_z_m
left_force_contact_gt10N
left_force_norm_N
phase_cos_times_pitch_deg
phase_sin_times_roll_deg
right_ankle_hip_x_m
right_ankle_hip_y_m
right_ankle_hip_z_m
right_force_contact_gt10N
right_force_norm_N
torso_pitch_deg
torso_roll_deg
continued / hidden / bc
continued / hidden / dagger_mixed
continued / input / bc
continued / input / dagger_mixed
scratch / hidden / bc
scratch / hidden / dagger_mixed
scratch / input / bc
scratch / input / dagger_mixed
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
R² after minus before (not percentage retained)


phase sin times roll deg
phase cos times pitch deg
future 200ms forward velocity mps
future 200ms roll deg
left force norm N
command forward displacement m
Held-out R²
Layer 1
Layer 2
Raw observations
Random layer 1
BC initial
BC matched
BC extended
DAgger initial
DAgger extended


Layer 1: geometry versus output improvement
Validation MSE reduction (%)
0
20
40
60
80
0.970
0.975
0.980
0.985
0.990
0.995
1.000
Mean within-run CKA to initializer
Layer 2: geometry versus output improvement
Validation MSE reduction (%)
0
20
40
60
80
0.93
0.94
0.95
0.96
0.97
0.98
0.99
1.00
Mean within-run CKA to initializer